
UNIT 8 MEASUREMENT AND SCALING TECHNIQUES

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8.0 OBJECTIVES

After going through this Unit, you will be able to:

- state the concept of measurement and its need;
- explain the various scales of measurement;
- discuss the various scaling techniques; and
- identify the criteria for good measurement.

8.1 INTRODUCTION

Measurement is the foundation of scientific enquiry. As we know that research begins with a 'problem' or topic. Thinking about a problem results in identifying concepts that capture the phenomenon being examined. Concepts are mental ideas representing the phenomenon. The process of conceptualization involves defining the concepts abstractly in theoretical terms. Operationalisation of concepts for research purpose involves moving from the abstract to the empirical level. This underlines the need to measure the various attributes of the people, the characteristics of objects or phenomenon. Further issues like decent work, human wellbeing, happiness, quality of education etc. which have several qualitative and quantitative dimensions are emerging important issues of research in economics. Such issues are being probed

through development of indicators and composite indexes which necessarily involve measurement of such indicators. Hence, a student of economics is expected to be well versed in the measurement scales, criteria of good measurement, and important scaling techniques. This unit throws light on these issues. Let us begin to discuss the concept of measurement.

8.2 CONCEPT OF MEASUREMENT

Simply speaking the process of assigning numbers to various attributes of people, objects or concepts is known as measurement. Technically, measurement is a process of mapping aspects of a domain to other aspects of a range according to some rule of correspondence. **Tyler(1963)** defines measurement as, “assignment of numerals according to rules”. **Nunnally (1970)** viewed that, “Measurement consists of rules, for assigning numbers to objects in such a way to represent quantities of attributes”. Thus, Nunnally focuses on both rules and manner in which numbers are assigned to an object.

According to **Campbell**, “Measurement is defined as the assignment of numerals to objects or events according to rules”. The fact that numerals can be assigned under different rules leads to different kinds of scales and different kinds of measurement.

Thus, measurement is a process to assign numbers or other symbols to characteristics of objects according to certain rules. Assigning numbers permit statistical analysis of the resulting data and facilitate the communication of measurement rules and results. The rules for assigning numbers hence need to be standardized and uniformly applicable. In the assignment process, there must be one to one correspondence between the numbers and characteristics being measured. In this process, variables can be divided into two basic types – quantitative and qualitative variables.

8.3 MEASUREMENT ISSUES IN RESEARCH

A number of concerns arise in the process of measurement which need to be addressed by the researcher. Some of the important issues include the following:

- 1) Whether the underlying characteristics of the concept allows ordering (ordinal level) or categorizing (nominal level)?
- 2) Whether the features of the concept are discrete or continuous with fine gradation?

The answer of these two questions will enable to determine the level of measurement inherent to the concept. As a general rule, efforts should be made that measurement represent the highest scale of measurement for a concept because it allows the use of more powerful statistical techniques for analysis.

- 3) Another important issue pertains to number of indicators for measurement of a concept. Some simple concepts can be measured by one indicator whereas abstract concepts are measured with more than one indicators. How many indicators are appropriate to measure a concept is to be decided by the researcher.
- 4) Another measurement issue concerns the source of valid and reliable measure. Hence, considerable attention should be given to identify valid and reliable measures for the study.

- 5) Measurement should be free from measurement errors like invalidity error and unreliability.
- 6) Measurement validity is distinct from internal validity and external validity because these two are separate research design issues.
- 7) A final measurement issue concerns the proper use of available data. For that, a research should be well aware of all available data sources and the types of data available with the various data compilation agencies.

8.4 SCALES OF MEASUREMENT

Before discussing different levels of measurement, let us remember that there are three postulates of measurement: (i) Equalities or identities, (ii) Rank order, (iii) Additivity.

You were introduced the elementary concept and types of scales of measurement in Unit 1. Even at the cost of repetition let us recall that Stanley S. Stevens (1946) in his seminal paper, "On the theory of scales of measurement" published in "Science" classified types of scales into four categories: (1) Nominal, (2) Ordinal, (3) Interval, (4) Ratio.

1) Nominal Scale

A qualitative scale without order is called nominal scale. In nominal scale numbers are used to name identity or classify persons, objects, groups, gender, industry type. Even if we assign unique numbers to each value, the numbers do not really mean anything. This scale neither has any specific order nor it has any value. In case of nominal measurement, statistical analysis is attempted in terms of counting or frequency, percentage, proportion, mode or coefficient of contingency. Addition, subtraction, multiplication and division are not possible under this scale.

2) Ordinal scale

A qualitative scale with order is called ordinal scale. In ordinal scale numbers denote the rank order of the objects or the individuals. Numbers are arranged from highest to lowest or lowest to highest. For example, students may be ranked 1st, 2nd and 3rd in terms of their academic achievement. The statistical operations which can be applied in ordinal measurement are median, percentiles and rank correlation coefficient. Ordinal scales do not provide information about the relative strength of ranking. This scale does not convey that the distance between the different rank values is equal. Ordinal scales are not equal interval measurement. Further this scale does not incorporate absolute zero point.

3) Interval scale

Interval scale includes all the characteristics of the nominal and ordinal scale of measurement. In interval scale numerically equal distance on the scale indicate equal distances in the attributes of the object being measured. For example, a scale represent marks of students using the attributes range 0 to 10, 10 to 20, 20 to 30, 30 to 40 and 40 to 50, and so forth. The mid point of each range (ie. 5, 15, 25, 35 and 45 etc.) are equidistance from each other. The data obtained from an interval scale is known as interval data. The appropriate measures used in this scale are: arithmetic mean, standard deviation, Karl Pearson's coefficient of correlation and tests like t-test and f-test. We cannot apply coefficient of variation in the interval scale.

4) Ratio Scale

Ratio scale has all the characteristics of nominal, ordinal and interval scale. It also possesses a conceptually meaningful zero point in which there is a total absence of the characteristic being measured. Ratio scales are common among physical sciences rather than among social sciences. The examples of ratio scales are the measures of height, weight, distance and so on. All measures of central tendencies that can be used in this scale include geometric and harmonic means.

The properties of these four scales can be summarized in following tabular form:

Properties Scale	Category	Ranking	Equal interval	Zero Point
Nominal	✓			
Ordinal	✓	✓		
Interval	✓	✓	✓	
Ratio	✓	✓	✓	✓

Properties Scale	Indicates Difference	Indicates Direction of Difference	Indicates Amount of Difference	Absolute Zero
Nominal	×			
Ordinal	×	×		
Interval	×	×	×	
Ratio	×	×	×	×

You will notice in the above tables that only the ratio scale meets the criteria for all four properties of scales of measurement. Interval and Ratio scale data are sometimes referred to as parametric and Nominal and Ordinal data are referred to as non-parametric.

Examples of Scales of Measurement –

Scale	Example	Statistics
Nominal	Gender Yes-no Students roll number Objects/Groups	Frequency, percentage, proportion Mode, Coefficient of contingency, chi-square (χ^2)
Ordinal	Class Rank, Socio Economic status Academic Achievement	Median, percentile, rank, correlation, Range
Interval	Student grade point, Temperature, Calendar dates, Rating Scale	Mean, correlation, Range, Standard deviation, rank order variance, Karl pearson's correlation, t-test, f-test etc.
Ratio	Weight, Height, Salary, Frequency of buying a product	Mean, Median, Mode, Range, Variance, Standard deviation, coefficient of variation, rank order variance, Karl pearson's correlation, t-test, f-test

8.5 CRITERION FOR GOOD MEASUREMENT

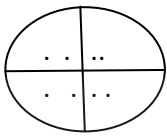
There are three major criteria for evaluating measurement:

- 1) Reliability
- 2) Validity
- 3) Practicality

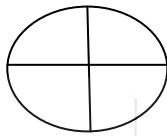
8.5.1 Reliability

Reliability refers to the degree to which the measurement or scale is consistent or dependable. If we use same construct again and again for measurement, it would lead to same conclusion. Reliability is consistency in drawing conclusion.

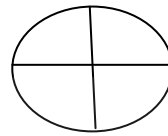
Reliability and Validity



Reliable and Valid



Valid but not Reliable



Reliable but not valid

8.5.2 Validity

Validity refers to the extent an instrument or scale tests or measures what it intends to measure. This means validity is the extent to which differences found with a measuring instrument reflect true differences among those being tested. This can be done by seeking other relevant evidences. There are three types of validity: content validity; criterion validity and construct validity. Content validity indicates the extent to which it provides coverage of the issues under study. Criterion validity examines how well a given measure relates to one or more external criterion based on empirical observations. And, construct validity explains the variation observed in tests conducted on several individuals. Construct validity is closely related with factor analysis.

8.5.3 Practicality

From practical view point, measure should be economical, convenient and interpretable. The measure should not be lengthy and difficult. The measure can be done by highly specialized persons.

8.6 ERRORS IN MEASUREMENTS

Measurement need to be precise and unambiguous for validity and reliability of any research study. Hence, measurement should be either free from errors or have minimum error. Any measurement usually involves two types of errors:

- Measurement invalidity
- Unreliability

Measurement invalidity refers to the degree to which the measure incorrectly captures the concept.

Unreliability refers to inconsistency in what the measure produces under repeated uses. If any measure, on average give some score for a case on variable and gives other score, when it is used again, it is said to be unreliable.

The possible sources of error are:

- a) **Respondent** by not expressing strong negative feelings.
- b) **Situation** wherein interviewer and respondent are not in good rapport.
- c) **Measurer** – errors may also creep because of incorrect coding, faulty tabulations and statistical calculations or behavior or attitude of the interviewer.
- d) **Instrument** errors may also arise because of the defective measuring instrument.

Check Your Progress 1

- 1) What do you mean by measurement?

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- 2) Give examples of interval scale and ratio scale.

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- 3) How will you examine the validity of a measurement?

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- 4) Identify three measure concerns that need to be addressed in the process of measurement.

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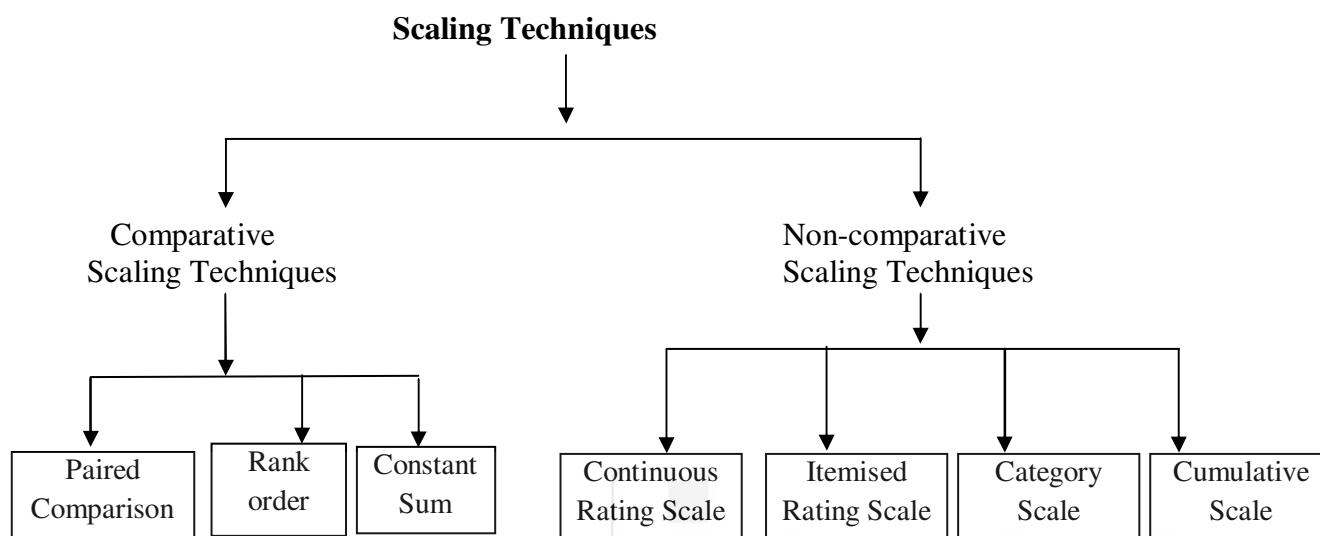
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8.7 SCALING TECHNIQUES

Scaling describes the way of generation of a continuum upon which measured objects are located. Broadly, scaling techniques can be classified into two categories: comparative, and non comparative scaling.



8.7.1 Comparative Scaling Techniques

Comparative scales involve the comparison of objects directly with one another. For example, in a study of consumer preferences for different telecom services, a respondent may be asked to rank choice according to his preferences. In this technique, data must be interpreted in relative terms and have only ordinal or rank order properties. In comparative scales small differences between stimulus objects can be detected.

Example

Rank the following factors in order of your importance in choosing mobile services (Assign 1 to most important and 5 to least important factor. Please do not repeat the ranks.)

Factor: Price, connectivity, call drop, Internet, Download speed.

Rank :.....

There are various comparative scaling techniques. Three most commonly used comparative scaling techniques are discussed here.

- a) Paired comparison
- b) Rank order
- c) Constant sum

a) Paired Comparison

The paired comparison is most commonly used scaling technique. This method is simply a binary choice. In this method respondents choose the stimulus or items in each pair that has the greater magnitude on the choice dimension they are instructed to use. This method is used when the study requires to distinguish between the two objects. The data obtained in this method is ordinal in nature.

Example

In a study of consumer preferences for the two brands of milk product i.e. Amul and Mother Dairy, a consumer is asked to indicate which one of the two brands he would prefer for personal use:

- 1) Which milk product of the following you prefer on the basis of taste. Please tick mark (✓).

Amul	Mother Dairy

- 2) Which milk product will you prefer on the basis of packaging? Please tick mark (✓).

Amul	Mother Dairy

- 3) Which product will you prefer on the basis of price? Please tick mark (✓).

Amul	Mother Dairy

This technique is useful when the researcher wants to compare two or more than two objects. If there are more than two objects (for example n objects) to compare, the total comparison will be;

$$\text{Number of comparison} = \frac{n[n-1]}{2};$$

n= number of objects.

b) Rank Order

This method is very popular among researchers and provides ordinal data. In this method, respondents are provided various objects and asked to rank the objects in the list. Rank order method is less time consuming. In this method, if there are n objects, only (n-1) decisions need to be made. Respondent can easily understand the instructions for ranking. This technique produces ordinal data.

Example

Rank the various brands of mobile phone in order of preference. The most preferred can be ranked 1, the next as 2 and so on. The least preferred will have the last rank. No two brands should receive the same rank number.

Sl. No.	Brand	Rank
1	Nokia	
2	Motorola	
3	Samsung	
4	HTC	
5	Karbons	
6	Spice	
7	Xolo	
8	Gionee	
9	LG	
10	Sony	

c) **Constant Sum**

In constant sum scaling, respondents are asked to allocate a constant sum of units to a specific set of objects on the basis of pre-defined criterion. If an object is not important, the respondent can allocate zero point and if an object is most important may allocate maximum points out of the fixed points. The total fixed points are 100. The total may be taken as some other value depending on the study.

Example

Allocate preferences regarding a movie based on various predefined criteria. Respondents were asked to rate each criteria in such a way that the total becomes 100.

Criteria	Respondent Preference
Content Quality	20
Music	20
Sound	15
Story	30
Breadth of Coverage (local, regional Supplements, national & global)	15
Total	100

The data obtained in this method is considered an ordinal scale. The advantage of this method is that it provides fine discrimination among objects without consuming too much time. However, allocation over large number of objects may create confusion for the respondent. This method cannot be used as a response strategy with illiterate people and children.

8.7.2 Non Comparative Scaling Techniques

Non-comparative scaling techniques involve scaling of objects independent to some specific standard. Respondent evaluate only one object at a time. For example, in a study of consumer preferences for different telecom services, a consumer may be asked to rate a list of factors that he/she would consider while choosing a particular telecom company.

Example

Please rate the following factors you think important in choosing mobile service. Rate 1 to least important and 5 to most important (rating could be on any scale. In this example, we have used a 5 point scale).

Factor	Price	Connectivity	Call Drop	Download Speed	Internet
Rating					

In this scaling technique, data is usually in interval scale. It can be continuous, metric/numeric also. Some of the commonly used non-comparative scaling techniques are –

- i) Continuous Rating Scale
- ii) Itemised Rating Scale
- iii) Category Scale
- iv) Cumulative Scale
- i) **Continuous Rating Scale**

This rating scale is also known as graphic rating scale. In continuous rating scale, respondents indicate their rating by marking at appropriate position on a line. The line is labeled at both ends from one extreme criterion to the other. The line may contain points 0,10,20,.....,100.

Example

How would you rate a magazine with regard to its quality.

Quality Indicators	Scale Measurement
Content Coverage	Most _____ x _____ Least Above 80 60 40 20 0
Language	Most _____ x _____ Least
Presentation Style	Most _____ x _____ Least

A very large number of ratings are possible if the respondents are literate to understand and accurately differentiate the objects. The data generated from continuous rating scale can be treated as numeric and interval data. The researcher can divide the line into as many categories as desired and assign scores based on the categories under which the ratings fall.